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Final Report

The Art of Scientific Computing: Project Title: Finite-Size Analysis of the Bak–Tang–Wiesenfeld Sandpile Model

Author:
Ziye Hu

Subject Handbook code:
COMP90072

Faculty of Science
The University of Melbourne

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Project expert: Jarra Horstman

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Chapter 1

Bak–Tang–Wiesenfeld Sandpile Model

1.1 Introduction

This project studies the Bak–Tang–Wiesenfeld (BTW) sandpile model [1, 2] on finite square lattices with open boundaries. In this model, one grain of sand is added at each time step. A site becomes unstable and topples when its height reaches four. This toppling can make neighbouring sites unstable and may produce a chain reaction called an avalanche.

The simulations examine how the sandpile develops from an initially empty lattice and approaches a statistically stationary state. Four quantities are recorded for each avalanche: the total number of toppling events, the number of distinct sites that topple (the avalanche area), the number of grains lost through the open boundary, and the maximum distance reached by the avalanche (the avalanche length). Lattice sizes from 3×3 to 20×20 are compared to investigate finite-size and boundary effects. The implementation is also checked using simple test cases, conservation of sand after accounting for boundary loss, and different valid toppling orders.

The model is useful because simple local rules can produce complex large-scale behaviour. Studying this model helps explain how a driven system can develop avalanches of many different sizes without requiring an externally adjusted critical parameter.

1.2 Theoretical background

1.2.1 Self-organised criticality

The BTW sandpile is a driven and dissipative system and is a standard example of self-organised criticality [1, 2]. It is driven because one grain of sand is repeatedly added to the lattice, and it is dissipative because grains can leave the lattice through the open boundary when an avalanche occurs. Sand is added slowly compared with the avalanche dynamics. A new grain is added only after the previous avalanche has completely stopped.

The system starts with an empty lattice. At first, the total amount of sand continues to increase. After an initial transient period, the average sand height no longer continues to rise. Instead, it fluctuates around a stable average value. The system has then reached a statistically stationary state. This is not a static state since grains are still being added, avalanches are still occurring, and sand is still leaving through the boundary.

The system is also not in equilibrium. It continuously receives sand from external driving and loses sand through the boundary. In the statistically stationary state, the average sand input is approximately equal to the average boundary loss.

1.2.2 Scale invariance and avalanche distributions

A scale-invariant system does not have one characteristic avalanche size. It can produce both small and large avalanches. A power-law distribution can be evidence of scale-invariant behaviour, although identifying a power law reliably requires more than observing an approximately straight line on a log–log plot [3]. A power law has the general form

$$P(x) \propto x^{-\alpha}, \tag{1.1}$$

where x is a measured avalanche quantity and α is an exponent. On a log–log plot, an ideal power law appears as a straight line.

For a finite lattice, the largest possible avalanche is limited by the lattice size. Therefore, the distribution has a finite-size cutoff. Larger lattices can support larger avalanches. For this reason, it is useful to compare several lattice sizes. In this project, the distributions are examined on logarithmic axes. However, an exact power-law exponent is not estimated.

1.2.3 The role of the boundary

Open boundaries allow sand to leave the lattice. When a corner site topples, two grains leave the lattice directly. When a non-corner edge site topples, one grain leaves directly. When an interior site topples, no sand leaves immediately. The boundary is therefore necessary for dissipation and affects the average height at different sites.

An infinite lattice has no boundary through which sand can escape. Therefore, a finite lattice with open boundaries behaves differently from an infinite system. As the lattice becomes larger, the fraction of sites located on the boundary becomes smaller. However, open boundaries are still needed in these simulations to remove sand from the system.

1.3 Model and method

The model uses square lattices with open boundaries. Four lattice sizes are considered: 3×3, 5×5, 10×10 and 20×20. Each site has an integer height z_{ij} representing its number of grains. Every simulation begins with an empty lattice.

At each time step, one grain is added to a uniformly selected random site. A site becomes unstable when its height reaches four. An unstable site topples according to

$$z_{ij} \to z_{ij} - 4. \tag{1.2}$$

One grain is transferred to each nearest neighbour. Grains directed outside the lattice leave the system. Toppling continues until all site heights are below four.

The same toppling rule can be written using a toppling matrix Δ. Its elements are

$$\Delta_{ij} = \begin{cases} 4, & i = j, \\ -1, & i \text{ and } j \text{ are nearest neighbours}, \\ 0, & \text{otherwise.} \end{cases} \tag{1.3}$$

A toppling at site i subtracts the corresponding row of Δ from the height vector. The diagonal elements are positive and the off-diagonal elements are non-positive. Interior rows sum to zero, while boundary rows have positive sums because grains can leave the lattice. This matrix form is also useful for explaining why valid toppling operations commute.

For each avalanche, the program records:

- **topples**: the total number of toppling events;
- **area**: the number of distinct sites that topple;
- **loss**: the number of grains lost through the boundary;
- **length**: the maximum Euclidean distance from the grain-addition site to any site that topples.

The 3×3, 5×5, 10×10 and 20×20 simulations use 10,000, 20,000, 50,000 and 100,000 grain additions, respectively. Their burn-in periods are 100, 500, 2,000 and 10,000 steps. Data from the burn-in period are excluded from the stationary analysis.

Each simulation is repeated using random seeds 42, 43, 44, 45 and 46. The reported comparisons use the mean and standard deviation across these five runs. Stationary mean heights are calculated using every time step after burn-in. Avalanche statistics are calculated only for time steps that produce at least one toppling event.

1.4 Validation

The original skeleton code was developed into a working `SandPile` class. The program can create a lattice, add grains, topple unstable sites, calculate boundary loss, and process a connected avalanche until the lattice is stable.

The implementation was tested using examples whose results could be calculated by hand. These included:

- toppling at the centre, an edge and a corner;
- a connected avalanche involving several sites;
- repeated toppling of the same site;
- two valid toppling orders applied to the same configuration;
- checks that the tested final configurations were stable; and
- checks of mass conservation.

For example, after 1,000 grain additions, one test simulation had a final mass of 16 and a total boundary loss of 984. Therefore,

$$M_{\text{input}} = M_{\text{final}} + M_{\text{loss}} = 16 + 984 = 1000. \tag{1.4}$$

The final program also performs an automatic mass-conservation check for every lattice size and random seed. All of these checks passed.

The two different valid toppling orders produced the same final stable configuration. This result is consistent with the Abelian property of the sandpile model, according to which the final stable configuration does not depend on the order of valid topplings [4, 5].

1.5 Results

1.5.1 Statistical stationarity

The average height initially increased because the simulations started from empty lattices. After the initial transient, the average height no longer showed a continuing upward trend. Instead, it fluctuated around a stable average value. An example for the 20×20 lattice is shown in Figure 1.1.

To check stationarity more directly, the data after burn-in were divided into three equal time blocks. For the 20×20 simulation with seed 42, the mean heights in these blocks were 2.06023, 2.06078 and 2.06068. Their close agreement shows that the mean height did not change systematically during the measured period.

The result was also consistent across independent random sequences. The stationary mean height of the 20×20 lattice, averaged across five random seeds, was

$$\langle h \rangle = 2.06084 \pm 0.00048, \tag{1.5}$$

where the value after the $\pm$ sign is the standard deviation across the five runs.

The mean boundary loss per stationary time step was approximately one grain for every lattice size. For the 20×20 lattice, the five-seed result was

$$\langle \text{loss per step} \rangle = 1.00004 \pm 0.00022. \tag{1.6}$$

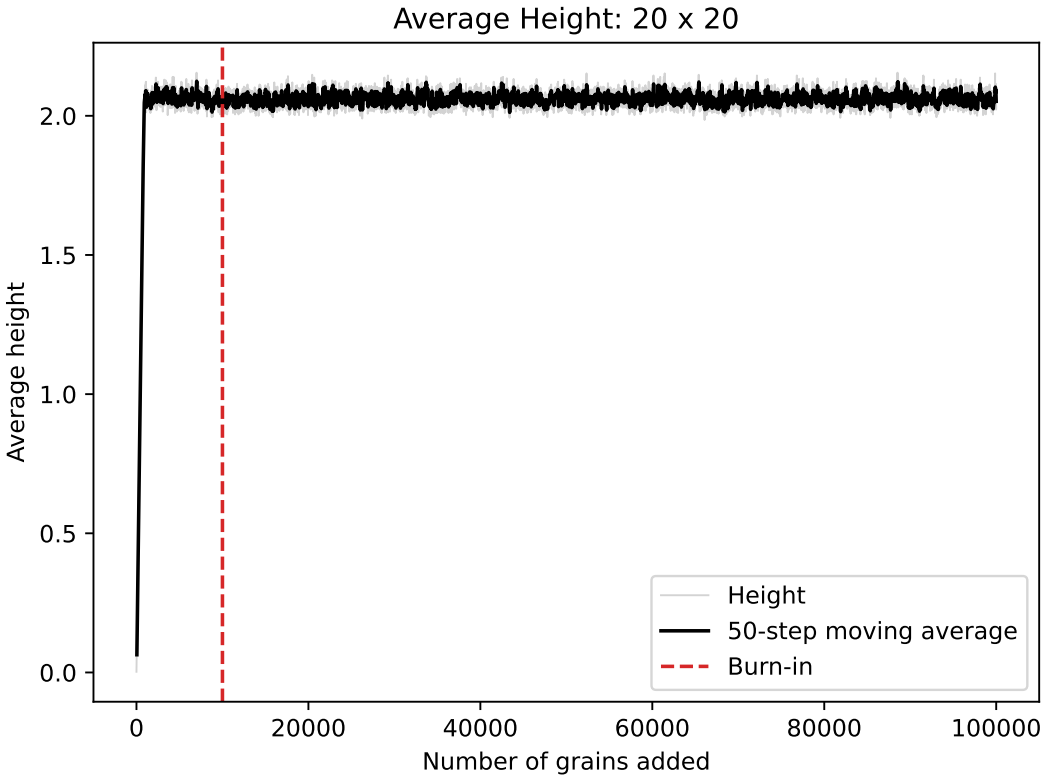


Figure 1.1: Mean lattice height during the 20×20 simulation after each grain addition. The red dashed line marks the end of the burn-in period at 10,000 grain additions. After the initial transient, the mean height fluctuates around a statistically stationary value of approximately 2.06.

This agrees with the input of one grain per time step. The stable mean height and the balance between average input and average loss show that the system reached a statistically stationary state.

However, this state is not static. Grains continue to be added, avalanches continue to occur, and grains continue to leave through the boundary. It is also not an equilibrium state because the system is continuously driven and dissipates sand. It is therefore more accurate to describe it as a statistically stationary non-equilibrium state.

1.5.2 Finite-size behaviour

The stationary behaviour changed systematically with lattice size. Figure 1.2 compares the results for the four lattice sizes. Each point is the mean of five random seeds, and the error bars show the standard deviation across these runs.

The stationary mean height increased from 1.8184 for the 3×3 lattice to 2.0608 for the 20×20 lattice. The probability that one grain addition produced an avalanche also increased, from 0.3216 to 0.4156. The small standard deviations show that these trends were consistent across the five random seeds.

The mean values of the four avalanche measurements are shown in Table 1.1.

Table 1.1: Mean avalanche measurements averaged across five random seeds. Only time steps that produced an avalanche are included.

L	Topples	Area	Loss	Length
3	2.547	2.542	3.109	0.876
5	4.712	4.659	2.831	1.514
10	12.812	12.158	2.557	2.838
20	40.812	35.830	2.406	5.047

The mean number of topples, avalanche area and avalanche length all increased with lattice size. Larger lattices provide more space for an avalanche to spread, so an avalanche can reach more sites and produce more toppling events.

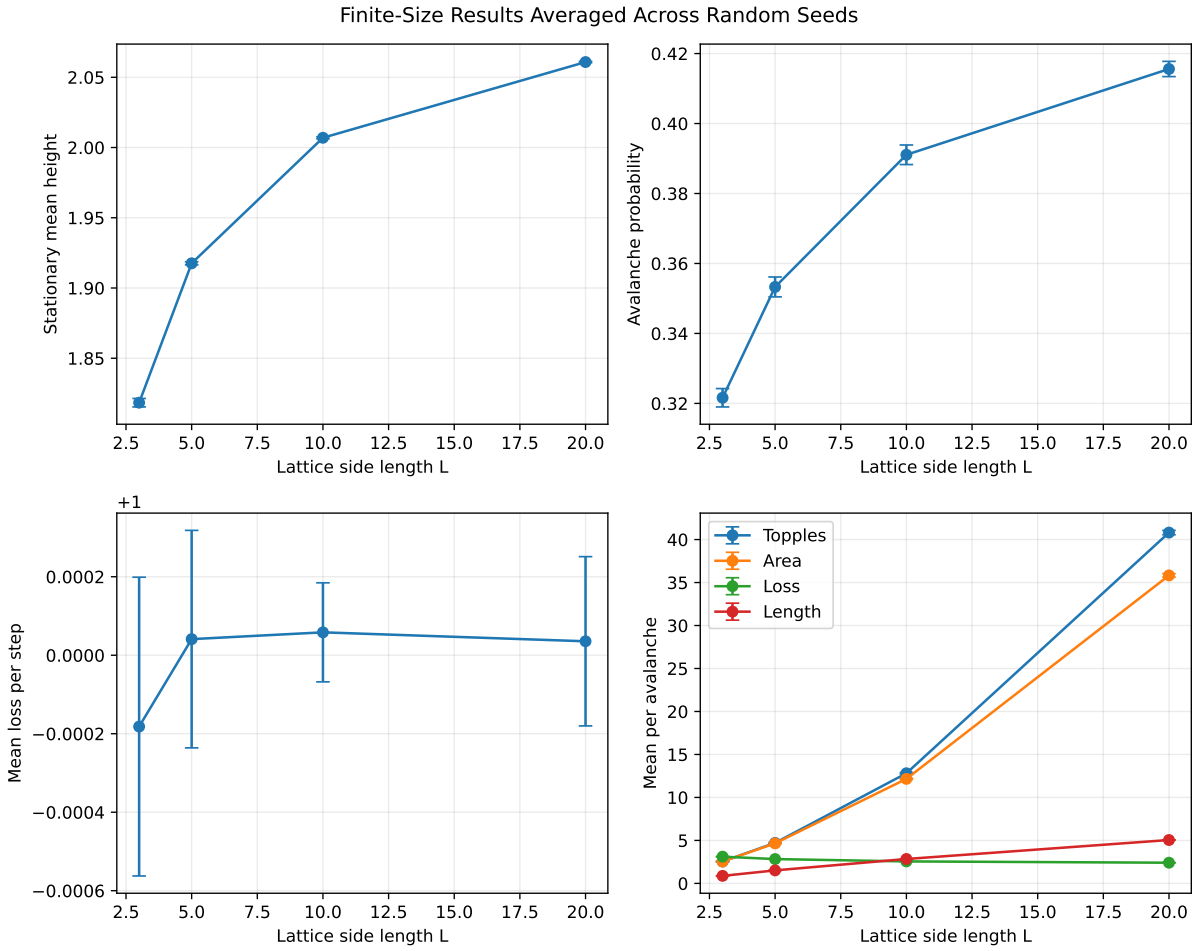


Figure 1.2: Comparison of statistically stationary measurements for lattice sizes from 3×3 to 20×20. Points show averages over random seeds 42–46, and error bars show one standard deviation between seeds. As lattice size increases, the stationary mean height, avalanche probability, mean number of topplings, avalanche area and avalanche length increase. In contrast, the mean boundary loss per avalanche decreases, while the mean loss per stationary step remains approximately one grain.

In contrast, the mean boundary loss per avalanche decreased from 3.109 to 2.406. On a small lattice, an avalanche can reach the boundary easily. On a larger lattice, many avalanches occur in the interior and stop before reaching the boundary. This reduces the mean loss for an individual avalanche.

This decrease does not mean that the larger systems lose less sand overall. Avalanches occur more frequently in the larger lattices, and the mean loss per time step remains approximately one grain. Therefore, the long-run boundary loss still balances the sand input.

1.5.3 Avalanche distributions

The distributions of topples, area, loss and length are compared in Figure 1.3. The data are shown on logarithmic axes and were grouped using logarithmic bins. Only time steps that produced an avalanche are included.

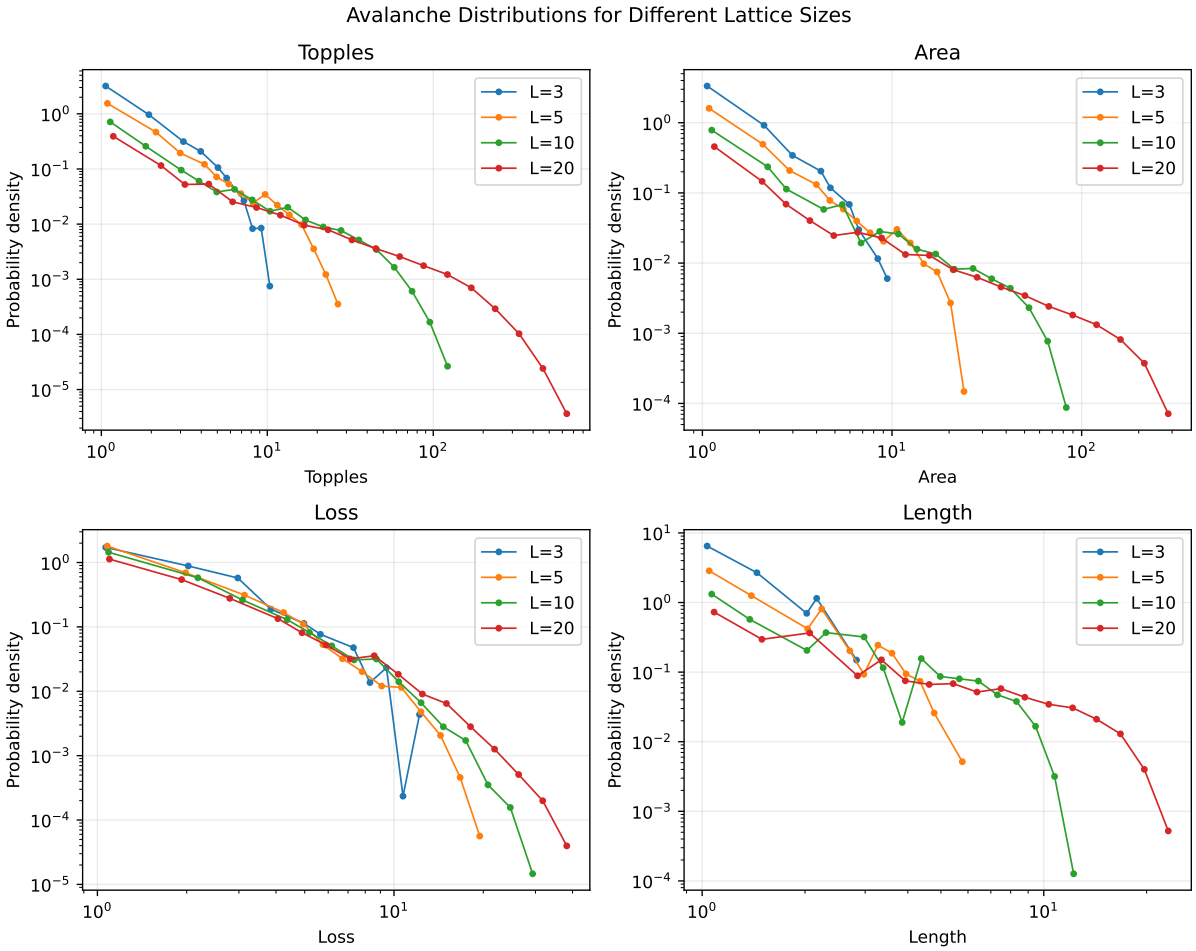


Figure 1.3: Logarithmically binned distributions of avalanche topples, area, boundary loss and length for lattice sizes from 3×3 to 20×20, using random seed 42. Small events occur more frequently than large events, while increasing lattice size extends the distributions towards larger values and produces size-dependent cutoffs. Zero values of loss and length are omitted because they cannot be displayed on logarithmic axes.

Small avalanches occurred more frequently than large avalanches for all four measurements. As the lattice size increased, the distributions of topples, area and length extended to larger values. This is consistent with the finite-size results because a larger lattice allows an avalanche to spread over a greater distance and involve more sites.

Each finite lattice also produced an upper cutoff in the avalanche measurements. The cutoff moved towards larger values as the lattice size increased. This indicates that the maximum observable avalanche size is limited by the size of the lattice.

The topples and area distributions have similar shapes because these measurements are strongly related. However, they are not identical. A site contributes only once to the avalanche area but can contribute several times to the number of topples. This difference becomes more important for larger avalanches, where repeated toppling is more common.

The loss and length distributions are less smooth. Boundary loss occurs only when an avalanche reaches an open boundary, while avalanche length can take only a limited set of distances on a finite square lattice. These effects produce more discrete and irregular distributions.

A power-law distribution would appear approximately as a straight line over a sufficiently wide range on a log–log plot. The present results show broad distributions and size-dependent cutoffs, which are consistent with finite-size critical behaviour. However, the available lattice sizes and data range are not sufficient to establish a precise power law or estimate a reliable exponent. Therefore, no exact power-law claim is made in this project.

1.5.4 Relationships between avalanche measurements

The relationships between topples, area, loss and length were examined using scatter plots and Pearson correlation coefficients. The six pairwise relationships for the 20×20 lattice are shown in Figure 1.4.

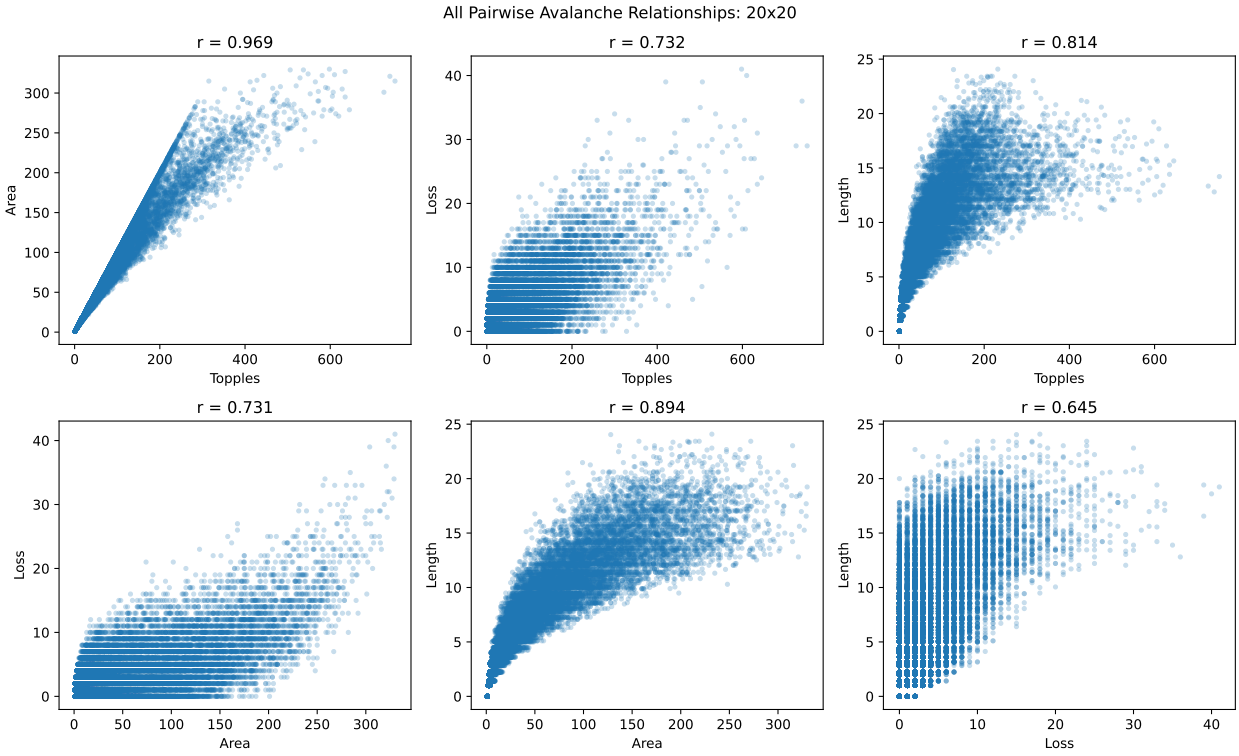


Figure 1.4: Pairwise relationships between topples, area, boundary loss and avalanche length for the 20×20 lattice using random seed 42. The value r in each panel is the Pearson correlation coefficient. Topples and area have the strongest relationship, while boundary loss has weaker relationships with the other measurements because it also depends on whether the avalanche reaches the open boundary.

All six correlation coefficients were positive. Their values for the 20×20 lattice are shown in Table 1.2.

Table 1.2: Pearson correlations between avalanche measurements for the 20×20 lattice using random seed 42.

Relationship	Correlation r
Topples–area	0.969
Topples–loss	0.732
Topples–length	0.814
Area–loss	0.731
Area–length	0.894
Loss–length	0.645

Topples and area had the strongest correlation. An avalanche that reaches more sites normally produces more toppling events. However, the correlation was not exactly one because the same site can topple more than once. Topples therefore increase faster than area in some large avalanches.

Area and length were also strongly correlated. An avalanche that travels farther from the addition site usually

reaches more distinct sites. The topples–length correlation was slightly weaker because an avalanche may produce repeated topplings without increasing its maximum distance.

Boundary loss had weaker correlations with the other measurements. A large avalanche is more likely to reach an open boundary, but its loss also depends on where it begins and which boundary sites topple. Two avalanches with similar topples or area can therefore lose different numbers of grains.

These results show that the four measurements contain related but different information. Topples measures the total amount of toppling activity, area measures the number of affected sites, length measures the spatial extent, and loss measures the interaction between an avalanche and the open boundary.

1.5.5 Boundary effects

The open boundary produces a spatial pattern in the long-run mean height. Figure 1.5 shows the mean height at each site of the 20×20 lattice.

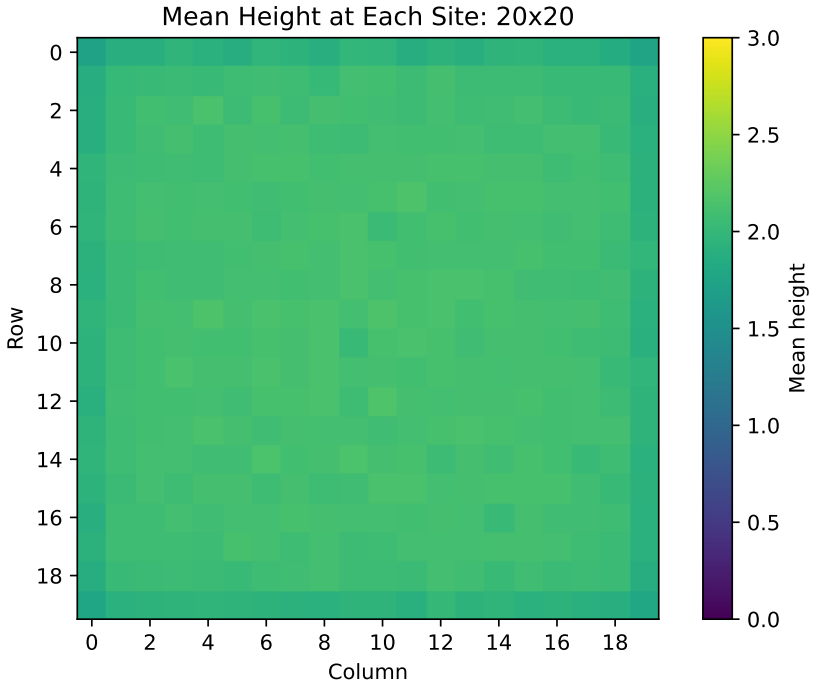


Figure 1.5: Mean height at each site of the 20×20 lattice after burn-in, using random seed 42. Interior sites have higher mean heights than edge and corner sites. This spatial pattern occurs because topplings at open boundaries dissipate grains directly, with corners producing the strongest local dissipation.

To compare different types of sites, the lattice was divided into corners, non-corner edge sites and interior sites. The results for random seed 42 are shown in Table 1.3.

Table 1.3: Mean heights at corner, non-corner edge and interior sites after burn-in, using random seed 42.

L	Corner	Edge	Interior
3	1.748	1.851	1.937
5	1.770	1.889	2.017
10	1.774	1.917	2.068
20	1.760	1.929	2.094

For every lattice size, corner sites had the lowest mean height and interior sites had the highest mean height. This pattern is caused by the open boundary. When a corner site topples, two grains leave the system directly. A non-corner edge site loses one grain directly, while an interior site has no direct boundary loss.

The difference between edge and interior heights became clearer for larger lattices. Interior sites can exchange grains with four neighbours without immediate dissipation. In contrast, boundary sites continually remove sand from the system. The open boundary therefore controls both the spatial height pattern and the long-run dissipation.

As the lattice size increases, the fraction of sites on the boundary decreases. This helps explain why the overall stationary mean height increased with lattice size. A larger fraction of the lattice is far from the dissipative boundary and can maintain a higher average height.

1.6 Discussion

1.6.1 Stationarity and self-organised behaviour

The simulations show the main features of a driven and dissipative sandpile. Sand is added slowly, unstable sites relax through avalanches, and grains leave through the open boundary. No external parameter is adjusted to place the system in a special state. Starting from an empty lattice, the system naturally approaches a regime with a stable mean height and continuing avalanche activity.

The statistically stationary state is maintained by a balance between driving and dissipation. One grain is added at every time step, and the measured mean loss per stationary step is approximately one grain. However, the instantaneous input and loss are not equal. Many additions produce no loss, while some avalanches lose several grains. The balance appears only after averaging over a sufficiently long period.

The stationary mean height increased with lattice size and reached approximately 2.061 for the 20×20 lattice. All stable site heights lie between zero and three, so a mean height slightly above two is reasonable. The results suggest that the mean height approaches a value near 2.1 as the lattice becomes large. However, the four lattice sizes used here are not sufficient for a precise infinite-lattice estimate.

1.6.2 Time and ensemble averages

Two types of averaging were used in this project. A time average was calculated from many steps of one simulation after burn-in. An ensemble average was calculated by repeating the simulation with five different random seeds.

The mean heights in the three stationary time blocks were very similar. The results from the five random seeds were also in close agreement. This shows that both averaging methods give consistent results for the observables studied here. This agreement is consistent with ergodic behaviour, where a sufficiently long time average represents the statistical behaviour of the system. However, a finite numerical simulation cannot provide a general mathematical proof of ergodicity.

1.6.3 Toppling order and the Abelian property

A toppling at site i changes the height vector by subtracting a fixed row of the toppling matrix. If T_i and T_j represent topplings at two sites, then

$$T_i T_j \mathbf{z} = \mathbf{z} - \Delta_i - \Delta_j = T_j T_i \mathbf{z}. \tag{1.7}$$

Therefore, two toppling operations commute whenever they are both performed. This is the Abelian property of the model.

The numerical validation showed that two different valid toppling orders gave the same stable configuration. More generally, the toppling matrix has positive diagonal elements, non-positive off-diagonal elements and non-negative row sums. Boundary rows have positive sums because they dissipate sand, and every interior site is connected to the boundary. These properties allow avalanches to terminate.

Because the toppling operations commute, changing the order of valid topplings changes only the intermediate configurations. It does not change the number of times each site eventually topples or the final stable configuration. The stabilization operation is therefore well-defined.

1.6.4 Interpretation of the avalanche measurements

The four avalanche measurements describe different parts of the same event. Topples measures the total amount of activity, while area counts the number of different sites involved. Length measures the maximum spatial extent, and loss measures how the avalanche interacts with the open boundary.

The strong relationship between topples and area is expected because an avalanche involving more sites normally requires more toppling events. Their difference becomes larger for large lattices because individual sites can topple repeatedly. Area and length are also strongly related, since avalanches that travel farther normally reach more sites.

Loss has weaker correlations with the other measurements because it depends strongly on the avalanche location. A large avalanche near the centre may stop before reaching the boundary, while a smaller avalanche starting near an edge may lose grains immediately. Boundary loss therefore cannot be predicted from avalanche size alone.

1.6.5 Limitations

The largest lattice studied was 20×20, so finite-size effects remain important. The avalanche distributions have clear upper cutoffs, and some measurements can take only a limited set of values. Longer simulations and larger lattices would provide a wider range for studying scaling behaviour.

The log–log distributions are compatible with broad avalanche statistics, but they do not prove an exact power law. A stronger claim would require a fitted exponent, uncertainty estimates, tests of the fitting range and comparisons with alternative distributions.

The results also apply specifically to uniform random grain addition, a toppling threshold of four and open square boundaries. Changing the driving rule, boundary condition or lattice geometry could produce different stationary and avalanche behaviour.

1.6.6 Possible extensions

This project concentrates on the baseline Bak–Tang–Wiesenfeld sandpile model and explores its finite-size properties, without modifying the underlying toppling rules. One valuable extension is to replace the open-square boundary condition with periodic boundaries along one axis to construct a cylindrical lattice. By comparing simulation outputs against those obtained from the original open-boundary model, one can quantify how the number and spatial arrangement of dissipative boundary sites influence observables including the stationary mean height, avalanche-induced grain loss and avalanche probability distributions.

A second promising modification is to deposit sand grains preferentially near the lattice centre instead of sampling drop sites uniformly across the whole domain. It would be anticipated that this driving scheme suppresses instances of immediate boundary dissipation, enabling avalanches to propagate over larger spatial distances before encountering an open boundary.

1.7 Conclusion

This project developed and tested a numerical implementation of the Bak–Tang–Wiesenfeld sandpile model on finite square lattices with open boundaries. The program successfully processed connected avalanches, recorded topples, area, boundary loss and length, and produced stable final configurations. Hand-calculated tests, different valid toppling orders and automatic mass-conservation checks were used to validate the implementation.

The simulations showed that the sandpile approaches a statistically stationary non-equilibrium state. In this regime, the average height fluctuates around a stable value and the mean boundary loss per time step is approximately equal to the input of one grain. Results from five random seeds were in close agreement.

Increasing the lattice size increased the stationary mean height, avalanche probability, mean number of topples, avalanche area and avalanche length. The mean loss per individual avalanche decreased, although the mean loss per time step remained close to one. Corner and edge sites had lower mean heights than interior sites because of direct boundary dissipation.

The four avalanche measurements were positively correlated but described different properties of an avalanche. Topples and area had the strongest relationship, while boundary loss had weaker relationships with the other measurements because it also depended on the avalanche location.

The avalanche distributions were broad and had size-dependent cutoffs. These results are consistent with finite-size critical behaviour, but the simulations do not provide enough evidence to establish an exact power law. Larger lattices, longer simulations and formal distribution fitting would be useful extensions of this work.

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